Aliah University

Indian Constitution

UCCUGAU 03

0 Credits, Full Marks- 80

Answer any four questions. Each question carries equal marks.

4x20= 80 marks

Q1. Explain the historical significance of the Constitution of India.

Q2. Discuss about the salient features of the Indian Constitution.

Q3. Explain the scheme of fundamental rights and their importance as enshrined in the Constitution of India.

Q4. Explain the legal status of directive principles of state policy and their importance.

Q5. Write a note on the concept of fundamental duties.

Q6. Write a note on the position and powers of the President of India.

Q7. Briefly explain the process for making amendments to the Constitution of India.

Q8. Short Notes- *(Any two)*

a) National Emergency

b) State Emergency

c) Financial Emergency

d) Separation of Powers

ALIAH UNIVERSITY
DEPARTMENT OF MATHEMATICS AND STATISTICS
COURSE: ENGINEERING MATHEMATICS-II
COURSE CODE: MATUGBS03(CEN, MEN)

All symbols have their usual meaning

Time-3 Hours — Full Marks-80

Group-A

1. Answer any **FIVE** questions: $2 \times 5 = 10$

(a) Write the relation between operators Δ and E. Also, prove it.

(b) Define Absolute error and Relative error in context of numerical problem.

(c) Prove that $1 + \delta^2\mu^2 = (1 + \frac{\delta^2}{2})^2$.

(d) Define Lagrangian function.

(e) Find the missing term from the following table using operators:

X :	0	1	2	3	4
Y :	1	3	9	?	81.

(f) Find the Lagrange's interpolation polynomial which corresponds to the following data:

X :	-1	-2	2	4
Y :	-1	-9	11	69.

(g) Write down the Error formula in Trapezoidal rule and Simpsons's $\frac{1}{3}$ rule.

Group-B

Answer any **THREE** questions from **2-6**

2. (a) From the following table of values x and $f(x)$ determine the value of $f(21)$ using Newtons's backward interpolation formula:

x :	0	5	10	15	20
$f(x)$:	1.2	1.8	3.8	8.6	16.4.

(b) From the following table of values x and $f(x)$ determine the value of $f(0.65)$ using Newtons's backward interpolation formula:

x :	0.3	0.4	0.5	0.6	0.7
$f(x)$:	0.6179	0.6554	0.6915	0.7257	0.7580.

[5+5]

3. (a) Find the Lagranges's interpolating polynomial from the following data and find the value at $x = 4.2$:

x :	-1	0	2	5
$f(x)$:	9	5	3	15.

(b) From the following table of values of x and $f(x)$ obtain $f'(x)$ and $f''(x)$ for $x = 1.6$:

x :	1.0	1.1	1.2	1.3	1.4	1.5	1.6
$f(x)$:	1.2661	1.3262	1.3937	1.4693	1.5534	1.6647	1.7500.

[5+5]

4. (a) Evaluate approximately, by Trapezoidal rule, the integral $\int_0^1 (4x-3x^2)dx$, by taking $n = 10$. Compute also the exact integral and find the absolute and relative error.

   (b) Compute $\int_0^{10} \frac{dx}{1+x}$, using Simpson's $\frac{1}{3}$ rule, taking $h = 1.0$ and compare the results with the exact value. [5+5]

5. (a) Find the location of the positive roots of $x^3 - 9x + 1 = 0$, and evaluate the smallest one by bisection method correct upto two decimal places.

   (b) Compute by the method of iteration the positive root of the equation $x^2 - x - 0.1 = 0$, correct upto three significant figures. [5+5]

6. (a) Find the solution of the following differential equation by Euler's method for $x = 1$ by taking $h = 0.2$:
   $\frac{dy}{dx} = x + y, \quad y = 1$ when $x = 0$. Calculate upto four significant figures. Compare the results with the exact solution.

   (b) From the Taylor series for $y(x)$ upto 6 terms; find $y(0.1)$ and $y(0.2)$ correct to four decimal places if $y(x)$ satisfies
   $y' = xy + 1, \quad y(0) = 1.$ [5+5]

Group-C

Answer any **FOUR** questions from **7-13**

7. (a) Form the partial differential equation of all spheres of radius λ, having centre in the xy-plane.

   (b) Form the partial differential equation by eliminating the arbitrary constants a and b from $\log(az - 1) = x + ay + b$.

   (c) Form the partial differential equation by eliminating the arbitrary functions f and g from $z = f(x^2 - y) + g(x^2 + y)$. [3+3+4]

8. (a) Solve: $(x^2 + 2y^2)p - xyq = xz$.

   (b) Solve: $z(z^2 + xy)(px - qy) = x^4$. [5+5]

9. (a) Solve: $px(z - 2y^2) = (z - qy)(z - y^2 - 2x^3)$.

   (b) Solve: $(y^3x - 2x^4)p + (2y^4 - x^3y)q = 9z(x^3 - y^3)$. [5+5]

10. (a) Show that the equation $p^2 + q^2 = 1$ and $(p^2 + q^2)x = pz$ are compatible and solve them.

    (b) Find a complete integral of $z^2(p^2z^2 + q^2) = 1$. [5+5]

11. (a) Solve $(D^2+3DD'+2D'^2)z = x+y$, by expanding the particular integral in ascending powers of D as well as in ascending powers of D'.

    (b) Solve: $(D - D'^2)(D + 2D'^2)z = (y + 1)e^x$. [5+5]

12. (a) Solve: $(D - D' - 1)(D - D' - 2)z = \sin(2x + 3y)$.

    (b) Solve: $(D^2 - DD' - 2D'^2 + 2D + 2D')z = e^{2x+3y}$. [5+5]

13. (a) Solve: $\frac{\partial^2 z}{\partial x^2} - \frac{\partial^2 z}{\partial y^2} + \frac{\partial z}{\partial x} + 3\frac{\partial z}{\partial y} - 2z = e^{x-y} - x^2 y$.

(b) Find the particular integral of $(D^2 + DD' - 6D'^2)z = x^2 \sin(x+y)$. [5+5]

Student Roll No.____________________

ALIAH UNIVERSITY

B. Tech. Semester:III, Civil Engineering

Odd Semester Examination Jan-2022

Subject Code &Name: (CENUGPC03), Building Materials

Time:3 Hours **Total Marks: 80**

Instruction: **(1) Figures to the right indicate full marks.**
(2) Assume suitable data if required.
(3) Question-1 is compulsory.

Que:-1 Answer the following questions **2×10=20**

A What is a frog? State it's important in clay bricks.
B State the differences between dry process and wet process of cement manufacturing.
C Draw a neat sketch of cross section of good timber and show its various components
D Describe the procedure of painting of new wood work
E Write a short note on Alkali-aggregate reaction.
F Why is cement mortar more commonly used than other mortars?
G Explains the terms hydraulic lime and quick lime
H Describe compaction factor test.
I Write a note on Flemish bond with neat sketch.
J Define terms- Rise, Tread, Newel posts, Balusters

Que:-2 Attempt any four. **5×4=20**

A Describe briefly the tests to which bricks may be put before using them for engineering purposes.
B What is meant by seasoning of timber? What are its objective?
C Write a short notes on bulking of sand.
D What are initial and final setting times of cement? What is their important?
E Compare paints and varnishes.

Que:-3 Attempt any four **10×4=40**

A Discuss the process of burning of bricks in Bull's Trench Kiln with neat sketch
B What are the ingredients of Portland cement? State the function and limits of each of them
C Explain the defects of timber with neat sketches.
D How do you determine the Flakiness Index and Elongated Index? What is its significance?
E What is workability of concrete? How you will measure the workability of concrete?
F Enlist and explain the different types of stairs used in public and residential buildings with neat sketches.

Aliah University

Odd Semester Examination 2021-2022

Subject Code: CENUGPC02

Subject Name: Surveying and Geomatics

B. Tech Civil Engineering III Semester

Full Marks: 80 (8x10); **Time: 3 Hrs.**

Attempt All Questions

1. A 20m chain was found to be 25cm to long of chaining a distance of 2500m. it was found to be 30cm too long at the end of the day after chaining a total distance of 5400m. Find the true distance of the distance work.
2. What are the different steps involved in the image processing of satellite data? What is the fundamental reason behind the FCC creation?
3. Explain the significance of Electromagnetic Radiation (EMR) in remote sensing? Give an example that differentiates active and passive remote sensing systems.
4. Describe the spectral signature concepts. Identify, what are the characteristic of EMR interaction with soil and vegetation.
5. Find the correction for curvature and refraction for a distance of
   (a) 1000m (b) 1.3km (c) 5000m (d) 5.7km
6. If an agricultural area, with crops such as rice, became flooded, what do you think, which dataset is used for the monitoring and management of agricultural land? Explain the reasons for your answers based on your knowledge of how radar energy interacts with a target.
7. What is the difference between GIS and GPS, also mentioned are their application in sustainable natural resource management? Describe the advantage and disadvantages of the raster and vector dataset.
8. In an old map, a line AB was drawn to a magnetic bearing of $5^0 30'$ the magnetic declination at the time being 1^0 East. To what magnetic bearing should the line be set now if the present magnetic declination is $8^0 30'$ East.

ALIAH UNIVERSITY
Department of Civil Engineering

End Semester Examination
CENUGPC01 : (Mechanics of Material)

Time – 3 Hrs **Total Marks – 80**

(Important Instructions: Answer ***all*** *questions. All are of equal marks. Your answer should be brief and to the point)*

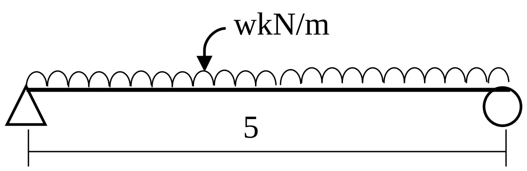

Fig. 1

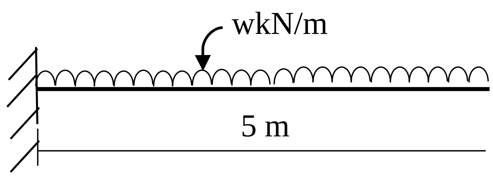

Fig. 2

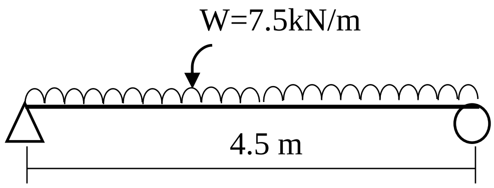

Fig. 3

1. Draw Shear force and bending moment diagram for the beams as shown in Fig -1 and Fig-2.
2. Derive the expression, $EI\frac{d^2y}{dx^2} = M$, where all the notations carry their usual meaning. Using this equation calculate the deflection at centre of the beam as shown in Fig-3.
3. Derive the stress-strain, strain- displacement, stress and strain compatibility relationships for the 2 D stress problem.
4. Derive the Euler's equation for the buckling of a column. Also provide an idea about the effective length of column.

Aliah University

Odd (3rd) Semester Examination, 2021-22

Course: Civil Engineering Year: Second Year Subject: Introduction to Civil Engineering

Subject Code: CENUGHUOI Max. Marks: 80 Time: 3 hrs

1. Answer any four questions: (20X4=80)

    a) What are the major streams of the Civil Engineering discipline? Briefly explain the major stream as per their application in the society.(5+15=20)
    b) What are the different types of foundation normally used in the construction sector. Briefly explain them as per their respective applicability to the type of construction. (5+15=20)
    c) Differentiate between concrete and mortar. What do you mean by workability of concrete. What is meant by characteristic compressive strength of concrete. What are the common types of bricks used in brickwork. (5+5+5+5=20)
    d) What are the considerations for selecting the tread and rise of a staircase? Differentiate between a dog legged staircase and a quarter turn staircase. What is meant by fineness, crushing strength, soundness and setting time of cement? Differentiate between mild steel and high carbon steel. What is meant by curing of concrete? Differentiate between artificial and natural seasoning of timber. (3+3+6+3+3+2=20)
    e) Briefly explain the general water supply scheme of an area through a flow chart. How is the total water quantity requirement for an area computed for preparing a water supply scheme of the same? What are meant by “Green Buildings”? What are the different ratings of “Green Buildings” as per the certifying authorities? Explain the importance of the selection of materials and orientation in designing a green building. (5+4+3+4+4=20)
    f) Explain the utmost importance of IS Codes in the discipline of Civil Engineering with reference to three relevant codes pertinent to three streams of Civil Engineering. Name some structures of civil engineering brilliance in and outside India. What is meant by the term superelevation related to highway engineering? What are the different types of sewers normally encountered? Differentiate between plinth level and ground level of a building. (5+4+3+4+4=20)

Estimation, Costing & Valuation(CENUGPC04) Assignment Submission Date:25-01-2022 To CR

1. Prepare an approximate estimate of the building project with total The plinth area of all buildings is 800 sqm. And from the following data.
   i) Plinth area rate Rs. 4500 per sqm
   ii) Cost of water supply @7½%of cost of building.
   iii) Cost of Sanitary and Electrical installations each @ 7½% of cost of
   Building.
   iv) Cost of architectural features @1% of building cost.
   v) Cost of roads and lawns @5% of building cost.
   Determine the total cost of the building project.

2. Prepare the rough estimate for a proposed commercial complex For a municipal corporation for the following data.
   Plinth Area = 500m2/floor
   Ht of each storey = 3.5m
   No.of storeys = G+2
   Cubical content rate = Rs. 1000/m3
   Provided for a following as a percentage of the structured cost
   a) water supply & Sanitary arrangement -8%
   b) Electrification -6%
   c) Fluctuation of rates - 5%
   d) Contractors profit - 10%
   e) Petty supervision & contingencies - 3%

3. To prepare the rough cost estimate of the Aliah University hostel building which accommodate 250 boys and 250 girl students. The construction cost, including all provisions, is Rs. 15,000/- per student. Determine the total cost of the building.

4. What are the Essentials of a contract? What is the type of contract? Explain valuation. What are the critical factors influencing the value of the building? What is the purpose of valuations? Write and explain the various methods of valuation.